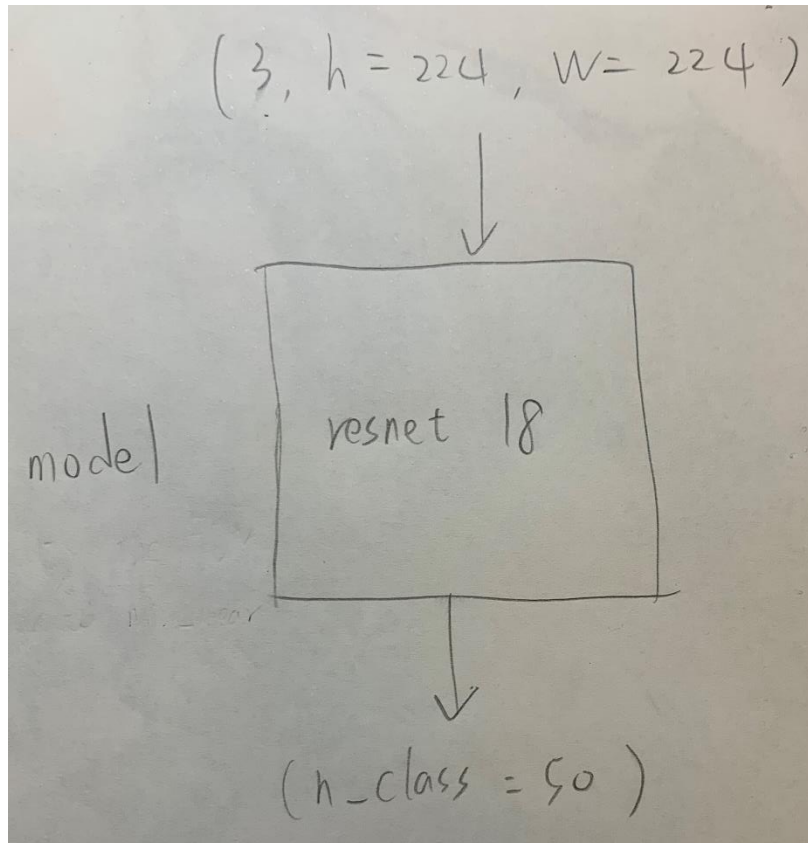


DLCV_hw1_report

r11942095 陳紹銘

Problem1

1.



2.

Model A:0.7393

Model B:0.8667

3.

The model A is ResNet18, so I construct a convolution block first. Then construct the entire model according the paper of ResNet and modify the output of the last layer(fully connected layer) to the number of classes of the dataset.

I do the data augmentation on training dataset and hope to get a better performance.

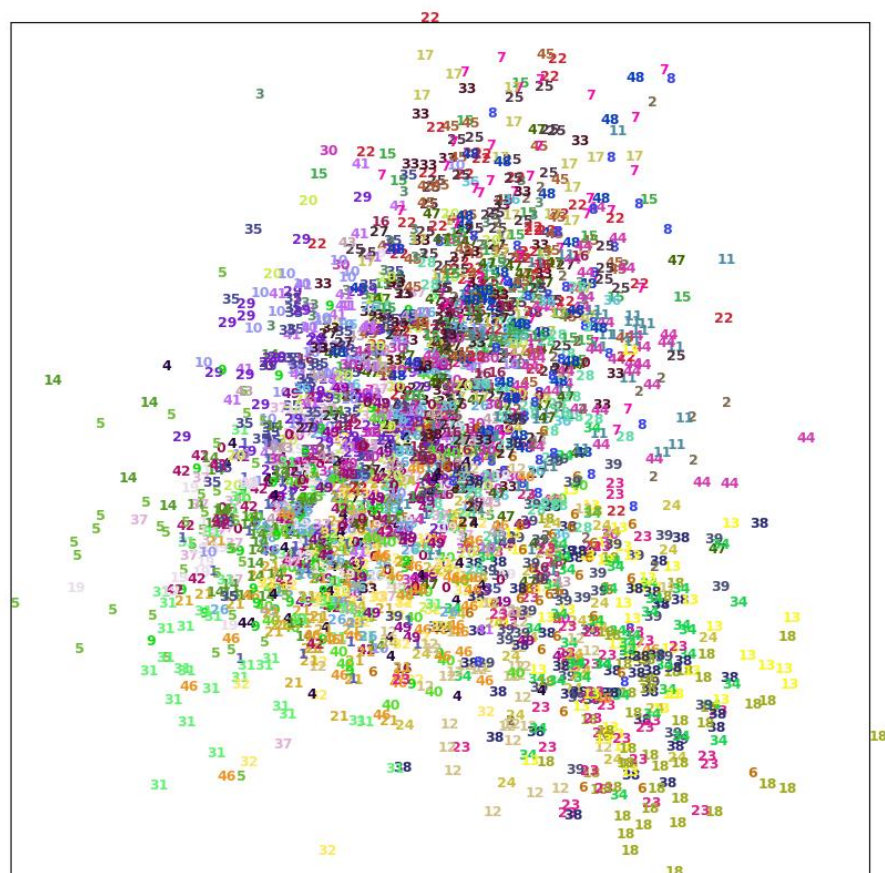
The loss function I used is cross-entropy loss.

The optimizer I used is SGD and I set the learning-rate to 0.01, momentum to 0.9, the rest parameter to the default value in Pytorch function.

4.

The model B I implemented is ResNet50. It has a deeper convolution than ResNet18(the model A) and so the entire model. Besides, I use the pretrained weight provided by Pytorch. Even though the pretrained weight is aimed to the ImageNet dataset, it still makes the model get 0.8 accuracy rate on validation set in 2 epochs.

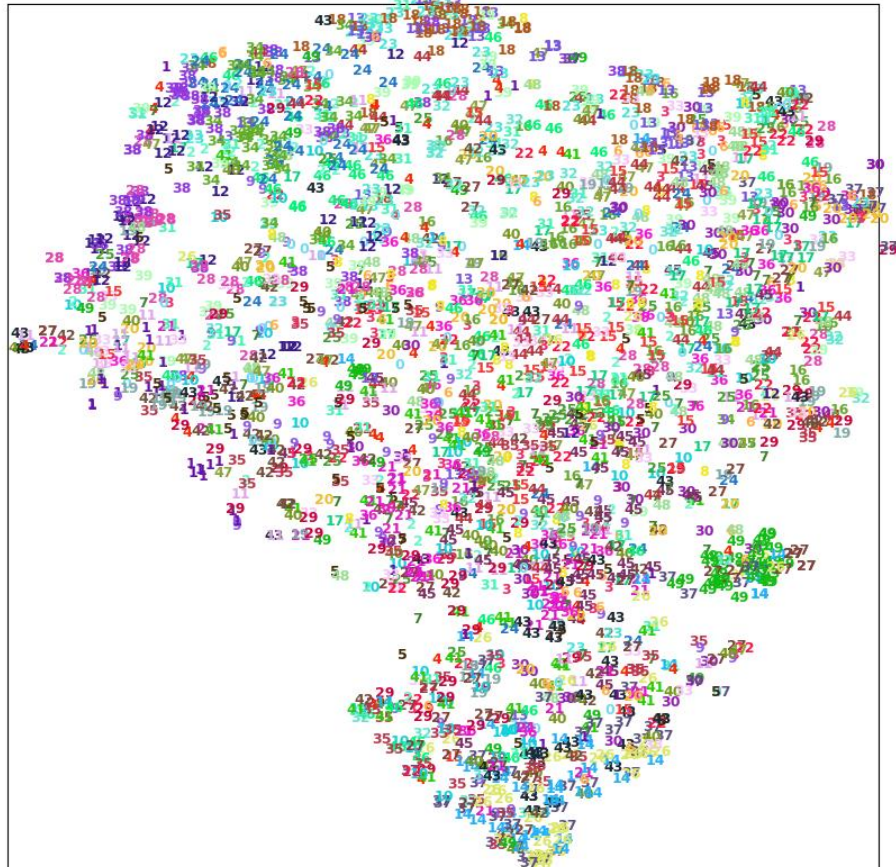
5.



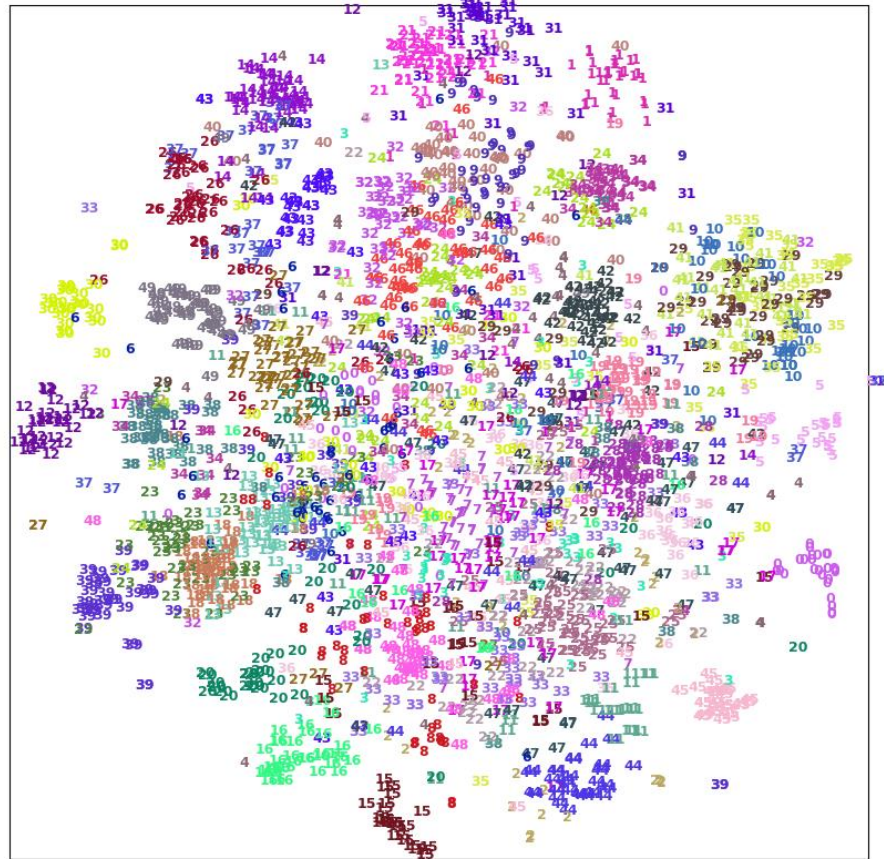
Since we have too many dimension(classes), so PCA still can't split the dataset properly, but the same label of images locate in nearby place.

6.

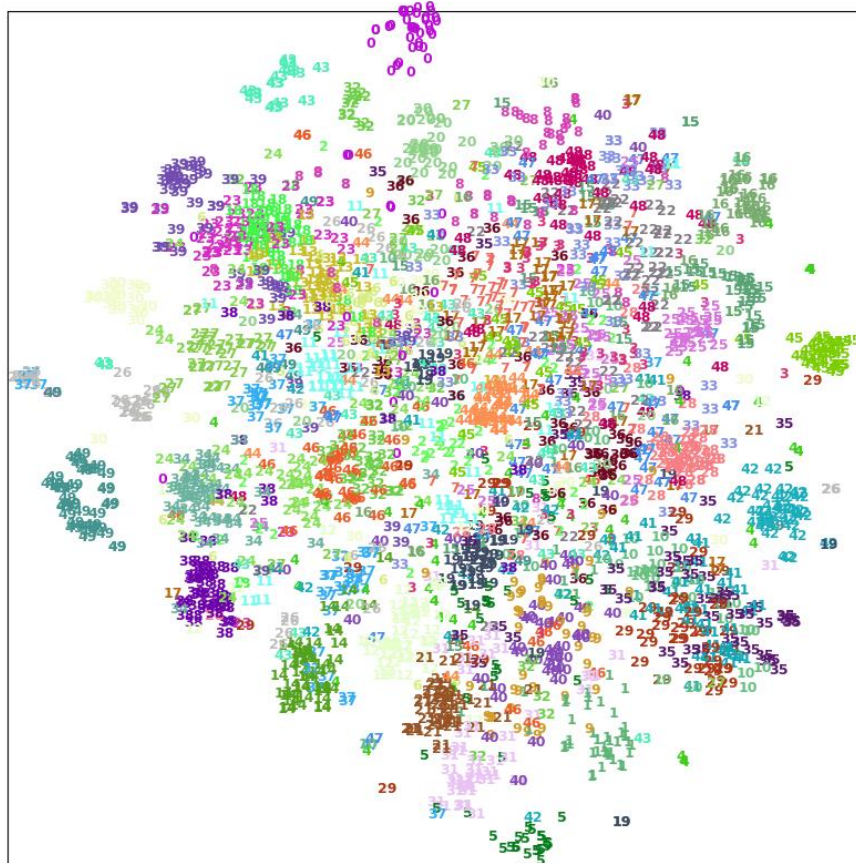
Epoch1:



Epoch50:



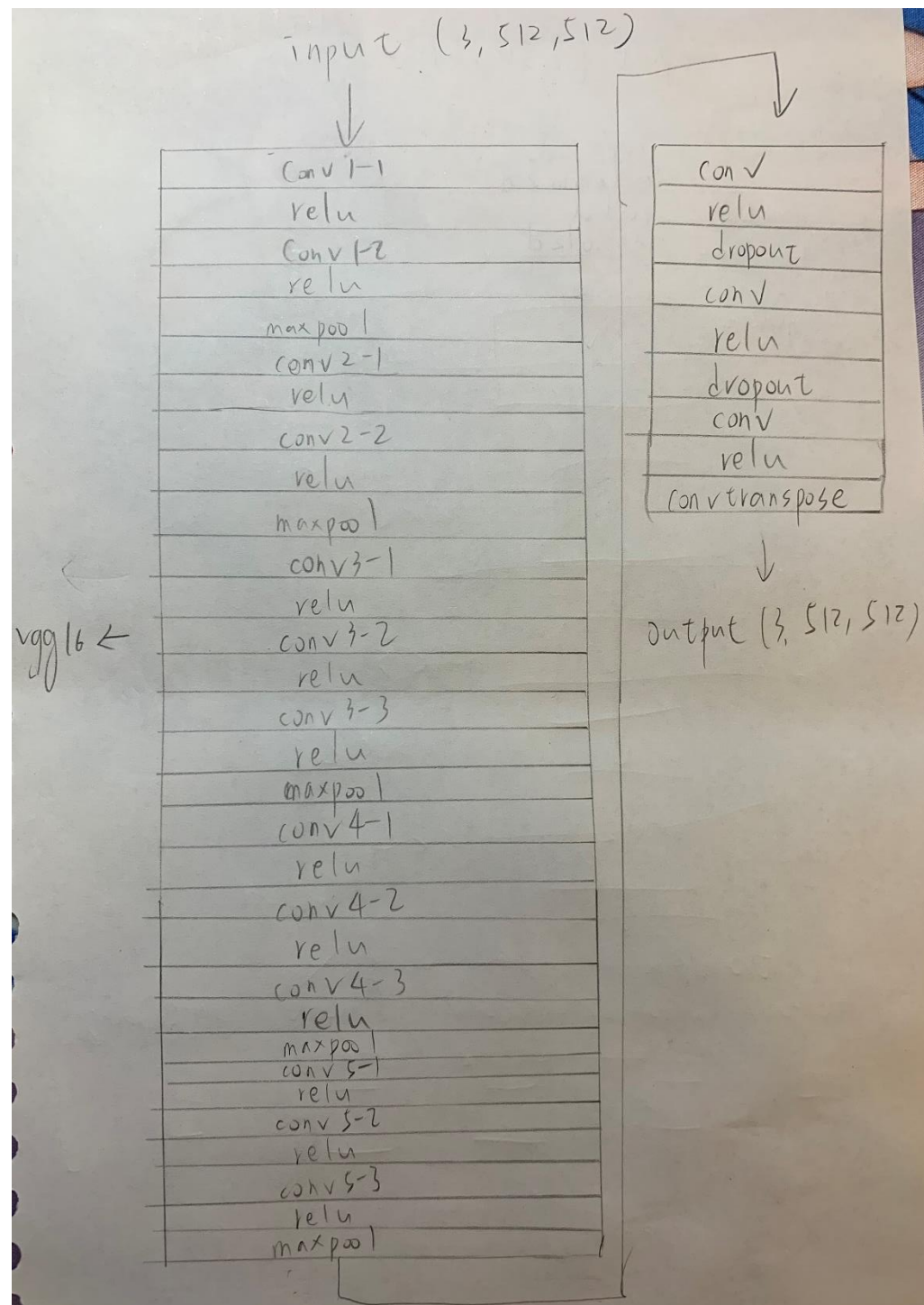
Epoch100:



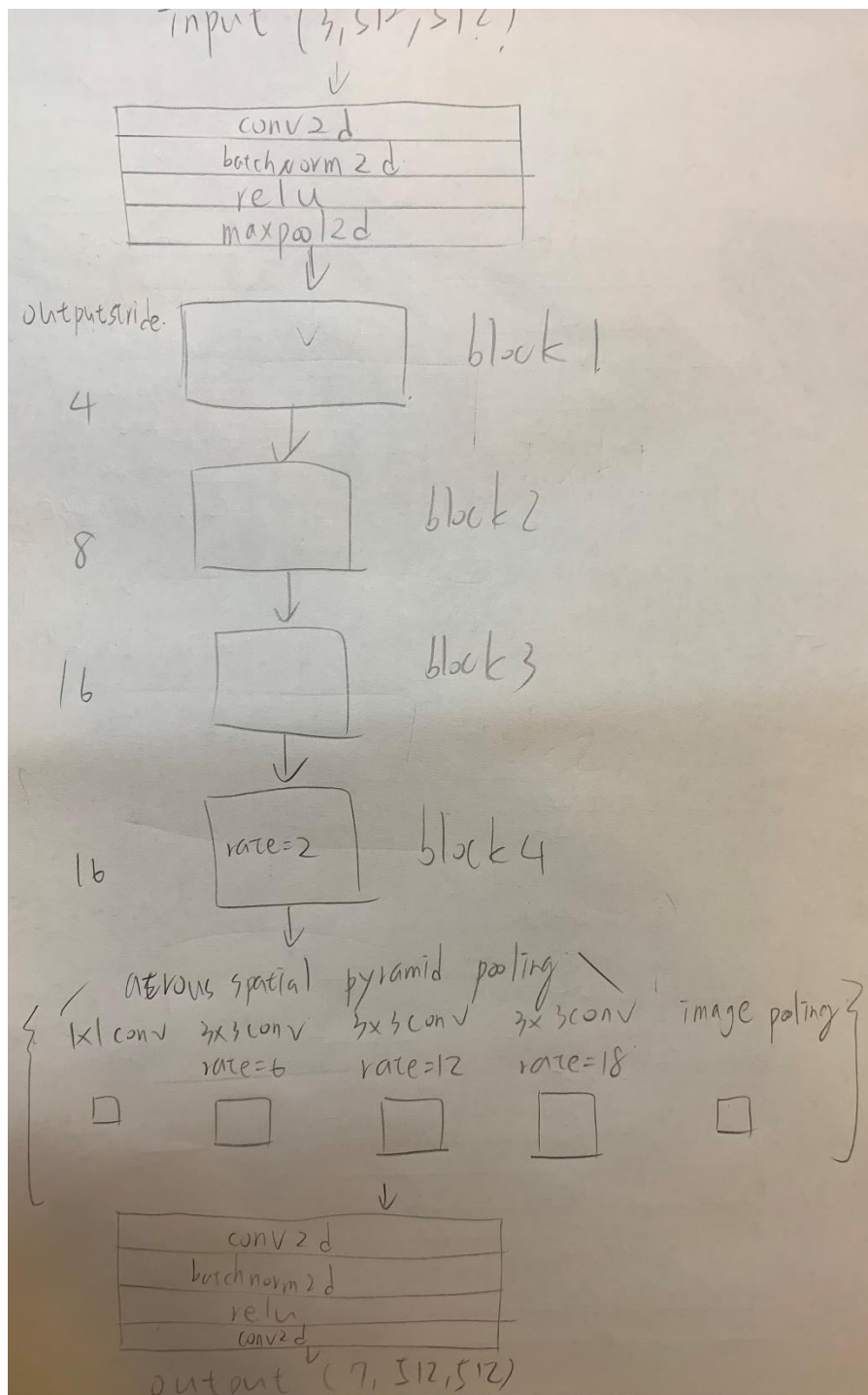
When the model is trained to the later stage, the same label of images get closer.
And the t-SNE performs the trend better than PCA as the image we show in 5.

Problem2

1.



2.



3.

ModelA:0.647

ModelB:0.728

4.

	Early(epoch1)	Middle(epoch12)	Final(epoch25)
0013_sat.png			
0062_sat.png			
0104_sat.png	